Benchmarking and Generating GP Consultation Summaries

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Abstract

This project explores how large language models (LLMs) can support clinical documentation by generating SOAP-structured notes from doctor–patient transcripts. We build an inference-only SOAP-generation pipeline and design an automatic benchmark, adapted from the FineSurE framework, to evaluate summaries on faithfulness, completeness, and conciseness. To assess how well such automatic scoring reflects expert judgment, we also propose a human evaluation study framework. Together, these components provide a structured approach for developing and validating LLM-based clinical summarization tools. The code is available under: https://github.com/Jokrasa1011/GP-summary-gen

1 Introduction

General practitioners (GPs) in the Netherlands experience increasing workload pressures, with consultation documentation forming a significant part of this administrative burden. Ambient scribing systems, which transcribe conversations and use large language models (LLMs) to generate structured notes, offer a promising way to reduce this effort. Because Dutch general practice relies on the SOAP format (Subjective, Objective, Assessment, Plan), generating accurate and clinically useful SOAP-notes requires both robust LLMs and reliable evaluation methods. In this project, we develop an LLM-based pipeline that converts doctor–patient transcripts into structured SOAP-notes. To assess the quality of these notes, we construct an automatic benchmark inspired by the FineSurE framework, designed to measure faithfulness, completeness, and conciseness at the level of each SOAP section. To understand how well this automatic evaluation reflects expert judgment, we design a human evaluation study that clinicians could use to rate the same summaries. Finally, we present our results, comparing model performance, analyzing benchmark behavior, and (potentially) examining the alignment between automatic and human assessment.

2 Related Work

As text summarization evolved, especially in the clinical domain, more sophisticated evaluation methods are needed. Traditional similarity metrics no longer suffice, particularly when dealing with high stakes applications like healthcare.

FineSurE as proposed by Song et al. (2024) moves away from Likert-scale ratings and towards more precise automated evaluation. The framework breaks assessment down into two main tasks. Sentence-level fact-checking that identifies specific error types and keyfact-alignment which verifies how much of the original information is retained and how much hallucinations extend the summary. This produces percentage scores for faithfulness, completeness and conciseness that correlate better with human judgment than earlier NLI or QA-based approaches.

UniSumEval tackles the problem of benchmarks for summaries typically drawing from limited domains and evaluating only some selected dimensions. Lee et al. (2024) instead proposes a benchmark covering nine domains, including medical literature. They use AI-assisted human labeling to get detailed assessments for the same three core dimensions that FineSurE also uses, faithfulness, completeness and conciseness. Notably, they included what they called "hallucinogenic texts" specifically to test whether LLMs produce a higher number of factual errors when operating in more difficult conditions.

MediNotes addresses the burden put on general practitioners of documenting electronic health records (EHR) by automating the process of SOAP-note-generation (Subjective, Objective, Assessment, Plan) from medical consultations. Leong

et al. (2024) achieved this by combining automatic speech recognition (ASR) with LLMs and retrieval-augmented generation (RAG). With those they process the conversational voice inputs into structured clinical notes in real time. The proposed system uses Quantized Low-Rank Adaption (QLorRA) and Parameter-Efficient Fine-Tuning (PEFT), making it more efficient, especially for medical institutes with limited computational resources.

Our Contribution builds upon these frameworks to create a domain-specific evaluation framework specifically tailored to GP consultation summaries. FineSurE could not be utilized as is because its default configuration, limited generic prompts, fails to capture the complexity and length of a complete clinical consultation. We therefore inject SOAP-specific context into the prompts to ensure facts are not just present, but correctly categorized. The summary evaluation itself is also split into each SOAP-section for a more detailed analysis. Furthermore, while MediNotes is an efficient generation pipeline, it does not provide the granular, section-specific benchmarking methods needed here. Similarly to the stress-testing methodology from Lee et al. (2024) using hallucinogenic texts specifically tailored to produce flawed outputs, we also tested with texts that intentionally include wrong information to verify the benchmark's ability to detect hallucinations in high-failure scenarios.

3 SOAP-Note Generation Pipeline

We use LLMs to automatically generate SOAP notes from consultation transcripts. In the prompt, we provide the model with the evaluation criteria, completeness, conciseness and faithfulness, as well as the expected content for each section of the SOAP note. We implement two generation strategies: a unified and split approach.

In the unified approach, a single prompt describes the complete SOAP note including the input we want for each section in one request. This method provides the LLM with broader context and allows it to better understand how the sections relate to each other. It makes it less likely that the same information is mentioned multiple times in different sections. However, it requires the model to maintain larger context windows and work with more complex instructions.

In the split approach, each section is generated through a separate request with section-specific instructions. Using this method, the model only has to generate a short paragraph and has specialized instructions for this specific part. This could lead to better adherence to the instructions. The trade-off is the loss of cross-sectional awareness of the whole SOAP note, which may lead to duplicates or inconsistent framing across sections.

4 Automatic Benchmark Design

An automatic benchmark for SOAP-note generation can be constructed by modifying and finetuning the general FineSurE methodology proposed by Song et al. (2024). While FineSurE is already multi-dimensional in terms of its output metrics (faithfulness, completeness, conciseness), its original implementation was designed for general text summarization and lacks the structural awareness required for clinical documentation. To address clinical documentation's strict accuracy requirements, we introduce a domain-specific adaptation that evaluates the automatically generated SOAP note both holistically and section-by-section using FineSurE's metrics. Furthermore, to ensure the robustness of our evaluation, we incorporate a stress-testing protocol inspired by the methodology of UniSumEval (Lee et al., 2024), utilizing hallucinogenic texts, in this case transcripts with specific distractors to verify a model's capabilities in error-prone scenarios.

4.1 Evaluation Design

The evaluation dimensions are faithfulness, completeness, and conciseness. The definitions given in FineSurE (Song et al., 2024) were adapted to SOAP-notes as follows.

- **Faithfulness**: A SOAP-note generator is faithful if each sentence in the generated note is fully supported or directly inferable from the source clinical record. A sentence is unfaithful when it introduces extrinsic content (hallucinations) or contains intrinsic distortions, such as incorrect entities, relations, temporal details, or predicate meanings. If faithfulness is fulfilled, the SOAP-note can be seen as factually correct on sentence-level.

- **Completeness**: A SOAP-note generator is complete if the generated note includes all clinically relevant key-facts identified in the source record, either directly or inferably from the output. This includes the reason for the visit, pertinent positive and negative findings,

vital signs, assessment elements, and plan actions.

- **Conciseness**: A SOAP-note generator is concise if the generated note contains only information that corresponds to clinically relevant key-facts and refrains from adding extraneous or unsupported content, yielding a succinct and focused representation of the source.

Based on these definitions, we compute the following scores, as defined in Song et al. (2024):

$$\text{Faithfulness } \% = \frac{\#\,\text{Correct Sentences}}{\#\,\text{Total Sentences}}, \quad (1)$$

$$\text{Completeness } \% = \frac{\#\,\text{Key-Facts Present}}{\#\,\text{Total Key-Facts}}, \quad (2)$$

$$\text{Conciseness } \% = \frac{\#\,\text{SA to } \geq 1 \text{ Key-Fact}}{\#\,\text{Total Sentences}}. \quad (3)$$

where SA stands for Sentences Aligned.

4.2 Section-Aware Evaluation Framework

FineSurE exists as a powerful tool that relies on existing LLMs to extract key-fact lists from the text and from the LLM generated summary. The original framework also only evaluates summaries as a single monolithic block of text. However, clinical SOAP-notes follow a strict structure where a fact can be valid in the *Subjective* section but may be considered a hallucination if placed in the *Objective* section. To address this, a structural parser is used to segment the generated SOAP note into its four constituent categories: Subjective, Objective, Assessment and Plan. The proposed split approach framework retains the the same fact checking as FineSurE, where a different LLM is used to compare the SOAP-note with the consultation transcript. Additionally, the default FineSurE prompts for the LLMs queries lack context that would assist with understanding key-facts linked to specific SOAP-sections. Therefore, enriching the prompts with specific context sentences allows the framework to properly link key-facts to the relevant dimension.

Our adapted framework performs evaluation at two levels of granularity:

1. **Section-Level Evaluation**: Each section is evaluated independently against a subset of key-facts specific to that domain. For example, the *Plan* section is evaluated solely against key-facts regarding medications, referrals, and follow-ups. This prevents the model from being penalized for missing information that belongs in a different section.

2. **Context-Enhanced Prompting**: We redesigned the LLM prompts to include medical context. Unlike the generic prompts in FineSurE, our prompts explicitly instruct the evaluator to check for medical error types, such as *entity errors* (wrong dosage/side), *predicate errors* (wrong diagnosis), and *circumstantial errors* (wrong timing).

This structural awareness allows pinpointing specific failure modes. For instance, a model that excels at capturing patient history (Subjective) but consistently hallucinates physical exam findings (Objective).

4.3 Aggregating Results

Individual section-level evaluation scores from each SOAP category must be combined into general, interpretable metrics to demonstrate the overall SOAP-note quality.

FineSurE metrics. SOAP category scores are computed independently using Equations 1,2,3, thus giving twelve intermediate evaluations which enables section specific performance analysis.

To bridge the gap from section-level scores to document-level scores, the results are aggregated using the geometric mean. The most intuitive option would be to take the arithmetic mean. This would result in rewarding senseless outputs for edge cases. For example, an empty output would only score 0% in completeness, however, since it is as short as possible and does not create any hallucination it would score 100% for both faithfulness as well as conciseness. An overall score of 66% should not be awarded for an empty output. The geometric mean mitigates this issue by disproportionately penalizing poor performance in any single dimension, thereby incentivizing consistency across all outputs. For a given consultation transcript T and a corresponding generated summary S, the overall scores are calculated as:

$$\text{Score}_m = \sqrt[4]{\prod_{c \in C} \text{Score}_{c,m}}$$
$$m \in \{\text{faithfulness, completeness, conciseness}\}$$
$$c \in \{\text{Subjective, Objective, Assessment, Plan}\} \quad (4)$$

where $\text{Score}_{c,m}$ represents the percentage score a given summary S obtained for a specific SOAP category and a given evaluation metric from the FineSurE framework. This aggregation ensures a balanced final benchmark for the entire medical note, though alternative weighting schemes could be justified for specific contexts where certain sections have more importance.

These document-level scores serve as the primary metric for later comparison between different models and approaches to the SOAP-note generation. Though all three metrics are of importance, in the realm of clinical application, faithfulness and completeness should be preferred for model comparison according to (Abacha et al., 2023).

General Performance Metrics. Besides document-level aggregation of the given FineSurE measures other metrics such as the standard deviation for a given model is also reported. This follows the practices in summarization evaluation (Song et al., 2024; Lee et al., 2024), where variance metrics are used to distinguish models that perform consistently from others. Especially in clinical applications, consistent performance is paramount. A model that occasionally produces highly inaccurate notes poses a greater safety risk than a model with consistent average results, regardless of having the same mean performance.

Error distributions are tallied separately for each SOAP section to provide further insights beyond the document-level average. Especially for faithfulness evaluation, the frequency of each error type from the FineSurE framework is recorded, following the classification system from (Song et al., 2024). This more detailed analysis reveals systematic failure, such as whether a model mostly hallucinates diagnostic conclusions in the Assessment category or misrepresents measurements in the Objective section of the SOAP note.

The proposed metrics are computed in a cross-lingual setting independently for Dutch and English, using translated transcripts. As proposed by (Lee et al., 2024), parallel evaluation across multiple languages tests specifically whether a model capabilities generalize well or if they are over-fitted on language specifics. Such possible performance discrepancies between more then one language show the need for additional optimization for a truly general model for SOAP-note generation.

4.4 Evaluator Validation

As we do not have access to ground-truth evaluations of the created summaries, we use controlled perturbations and check whether the evaluation changes as expected. This is similar to the stress-testing methodology proposed in UniSumEval (Lee et al., 2024). We apply three perturbations to each SOAP note to test changes in the three scores(each created by a different LLM). We compare these changes to the expected change. Let N_s be the number of sentences in the section, N_k the number of key-facts, N_a the number of sentences aligned to at least one key-fact, and F, C, Q the section-level faithfulness, completeness, and conciseness scores as defined in Equations 1, 2 and 3.

- **Omission - Completeness:** remove two sentences per section such that two key facts are no longer contained in the summary. Expected change of completeness score: $\Delta C = \frac{(C \cdot N_k - 2)}{N_k} - C$.
- **Irrelevant addition - Conciseness:** add two sentences per section that are not relevant. Expected change of conciseness score: $\Delta Q = \frac{Q \cdot N_s}{N_s + 2} - Q$.
- **Incorrect addition - Faithfulness:** add two sentences per section that contradict the facts. Expected change of faithfulness score: $\Delta F = \left(1 - \frac{(1-F) \cdot N_s + 2}{N_s + 2}\right) - F$.

We then use the mean squared difference between the actual changes and the expected changes as well as the rate of negative changes as a proxy for evaluator quality. An optimal evaluator would have a mean squared difference of 0 and a negative change rate of 100%.

5 Human Expert Study

To ensure consistency between the automatic and human evaluations, we retain the definitions of faithfulness, completeness and conciseness as stated in *Evaluation Design* (Section 4.1). Human experts are tasked with assessing SOAP-notes using the identical criteria to allow for direct comparison against the automatic benchmarks.

5.1 Data Collection Protocol

Following UniSumEval's multi-level annotation methodology (Lee et al., 2024), each section of a SOAP-note is independently evaluated by at least

three clinical annotators (licensed clinicians or senior medical students), which improves reliability and allows computation of inter-annotator agreement (IAA) metrics (Pagnoni et al., 2021; Bhandari, 2020).

Human evaluators complete two structured tasks (Appendix A):

Fact Checking At the sentence level, annotators identify the presence of factual errors (faithfulness) using a binary label. When errors are present, they are categorized using the seven FRANK error types (Pagnoni et al., 2021).

Key-Fact Alignment Annotators assess whether each key-fact is present in the corresponding SOAP note summary section, recorded as a binary decision. Completeness and conciseness are derived from these judgments, and majority voting is used to mitigate noisy annotations (Bhandari, 2020).

Binary decision schemes are used throughout, as they reduce ambiguity and improve inter-rater agreement compared to Likert scales (Lee et al., 2024). Annotators receive a detailed codebook, examples, and practice items, following UniSumEval's calibration strategy. Attention checks (20%) are included to ensure annotation quality. Annotators are compensated per summary, typically €0.20–€0.40 (Pagnoni et al., 2021; Bhandari, 2020). Given the complexity of long clinical inputs, AI assistance is recommended but not mandatory, including similarity-based sentence highlighting (top 30%) and pre-generated fact-checking suggestions that annotators can confirm or reject (Lee et al., 2024).

5.2 Inter-Annotator Agreement

To assess annotation reliability, we measure inter-annotator agreement, which quantifies the consistency with which different annotators label the same data. High agreement indicates reliable annotations and clear guidelines, whereas low agreement suggests ambiguity in the task, instructions, or data.

We report the following agreement measures:

Krippendorff's α for key-fact alignment as suggested in REALSumm(Bhandari, 2020), which considers multiple annotators (Krippendorff, 2011). A good value for Krippendorff's α should be 0.8 or higher, values below 0.67 are not considered reliable (De Swert, 2012).

It is defined as:

$$\alpha = 1 - \frac{D_o}{D_e}, \tag{5}$$

where D_o denotes the observed disagreement and D_e the expected disagreement by chance. For nominal data, the disagreement function is:

$$\delta(c, c') = \begin{cases} 0, & \text{if } c = c', \\ 1, & \text{if } c \neq c', \end{cases} \tag{6}$$

indicating that only exact label matches contribute to agreement.

Gwet's AC1 for IAA in fact checking tasks, which provides a more stable chance-corrected agreement estimate and is less sensitive to prevalence and marginal probability than Cohen's κ (Wongpakaran et al., 2013; Lee et al., 2024). To use κ as an IAA for factual validation and error type categorization was also suggested in FRANK (Pagnoni et al., 2021). A good Gwet's AC1 is 0.7 though scores of 0.6 for complex tasks might be acceptable as well (Wongpakaran et al., 2013). It is computed as:

$$\text{AC1} = \frac{P_o - P_e}{1 - P_e}, \tag{7}$$

where P_o is the observed agreement and P_e is the chance agreement.

5.3 Alignment between Human Evaluation and Benchmark

To quantify alignment between the benchmark and human-derived quality metrics, we can employ multiple statistical methods validated in FineSurE (Song et al., 2024) and UniSumEval (Lee et al., 2024):

SOAP-note-level correlations We can compute both Pearson's ρ (linear consistency) and Spearman's r_s (rank consistency) correlation coefficients to assess alignment regarding all three evaluation dimensions on summary-level, following the correlation analyses conducted in (Song et al., 2024; Lee et al., 2024).

$$\rho = \frac{cov(X, Y)}{\sigma_X \sigma_Y} \tag{8}$$

where cov is the covariance, σ_X is the standard deviation of X and σ_Y is the standard deviation of Y.

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{9}$$

Table 1: SOAP Generation Results with Standard Deviation

Model	Faithfulness	Completeness	Conciseness	Composite
deepseek-v3-2_unified	97.5±1.5	83.7±3.7	57.9±11.4	77.6±6.4
deepseek-v3-2_sectional	97.1±3.0	86.8±6.7	47.5±10.2	73.2±5.1
claude-opus-4-5_unified	97.4±1.2	84.5±11.1	52.4±13.2	75.1±8.4
claude-opus-4-5_sectional	97.6±1.8	89.8±4.8	45.8±12.1	73.3±7.1
gemini-2-5-pro-preview-06-05_unified	98.7±0.9	85.5±7.6	47.1±13.2	72.9±6.6
gemini-2-5-pro-preview-06-05_sectional	98.0±1.0	88.4±5.1	43.7±9.0	72.1±5.9
gemini-2-0-flash-001_unified	97.0±1.7	82.4±4.8	58.5±8.5	77.4±4.5
gemini-2-0-flash-001_sectional	98.0±1.7	83.5±6.5	44.9±11.7	71.1±7.2
gpt-oss-120b_unified	86.1±6.0	85.0±4.9	52.4±10.4	72.4±6.7
gpt-oss-120b_sectional	95.2±2.2	84.6±7.3	48.2±9.4	72.6±6.2
qwen3-next-80b-a3b-thinking_unified	97.4±3.3	45.0±41.3	78.2±21.4	46.0±39.6
qwen3-next-80b-a3b-thinking_sectional	83.1±22.5	44.6±31.6	41.1±20.9	51.3±26.5
gpt-5-nano_unified	99.4±1.8	8.9±28.2	96.6±10.8	9.1±25.6
gpt-5-nano_sectional	89.2±8.8	35.3±23.3	44.5±18.2	49.7±19.8

Sentence-level faithfulness For sentence-level error classifications, we compute balanced accuracy (bAcc), which is robust to skewed error distributions:

$$\text{bAcc} = \frac{1}{2}\left(\text{sensitivity} + \text{specificity}\right), \quad (10)$$

$$\text{bAcc} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right), \quad (11)$$

as recommended in FineSurE's sentence-level factuality evaluation (Song et al., 2024).

System-level ranking. If multiple models are evaluated, their average human and benchmark scores are compared using Spearman rank correlation at system-level, as implemented in UniSumEval's model-level analysis (Lee et al., 2024).

6 Experimental Setup

We run the generation pipeline across a diverse range of models, including both proprietary and open-source models of various sizes and reasoning capabilities: DeepSeek-V3-2, Claude Opus 4.5, Gemini 2.5 Pro, Gemini 2.0 Flash, GPT-OSS-120B, Qwen3-Next-80B-A3B-Thinking, and GPT 5 Nano. Each model is run using both the unified and the split prompting approaches. As the original dataset is in Dutch, we first use the Deepl API to translate it to English.

For evaluation, we employ multiple models to assess the robustness of evaluation scores across different evaluator models. However, since each evaluator must process outputs from all generator–approach combinations, we restrict evaluation to the computationally efficient models: GPT-5-Nano, Gemini 2.0 Flash and Qwen3-Next-80B-A3B-Thinking. Using larger frontier models such as Gemini 3.0 Pro, Opus 4.5 or GPT-5.2 would be too expensive. We average each model's score to calculate the total scores of each category (faithful, concise, complete) and then calculate the composite score based on that.

To evaluate generalization to other languages, we run our framework with a dutch dataset as well. We run it with a reduced model subset to reduce costs and runtime with Gemini 2.0 Flash, GPT 5 Nano and Qwen 3 Next 80b Thinking as both evaluator and generator models.

Evaluator validation is run on the perturbation subset with the same three evaluators (Gemini 2.0 Flash, Qwen3-Next-80B-A3B-Thinking, GPT-5-Nano) and section-level evaluation only. We manually apply the changes to the original summary. We use one summary per generation model, to get a variety of different styles.

The source code is available in this Github repository: `https://github.com/Jokrasa1011/GP-summary-gen`. Datasets and results are not included as they are not supposed to be shared publicly.

Evaluator	Omit (Comp.)		Irrel. (Conc.)		Incorrect (Faith.)	
	MSE	Neg. rate	MSE	Neg. rate	MSE	Neg. rate
anthropic/claude-opus-4.5	0.106	0.57	0.001	0.82	0.008	0.68
google/gemini-2.0-flash-001	0.132	0.54	0.001	0.82	0.026	0.79
openai/gpt-5-nano	0.106	0.64	0.006	0.68	0.006	0.68
qwen/qwen3-next-80b-a3b-thinking	0.223	0.54	0.054	0.71	0.085	0.68

Table 2: Evaluator perturbation sensitivity (section-level). MSE is mean squared error between actual and expected deltas; Neg. rate is the fraction of sections with negative delta.

7 Results

Table 1 shows the average mean scores of all models under both split and unified generation for each evaluation category. Larger and more recent models achieve higher scores than smaller, older models. While smaller models perform worse across all models, differences are especially visible in conciseness and completeness while faithfulness is often high as well. Unified prompting generally yields better results than split prompting for models that are strong already (Deepseek v3.2, Claude-Opus 4.5, Gemini 2.0 Flash and Gemini 2.5 Pro) while smaller models are stronger with the sectional approach (GPT 5 Nano, Qwen3 Next 80b). Even the strong models struggle to achieve high scores in conciseness.

The Dutch run achieves very similar scores to the English one, with the same ordering and similar differences between scores of the models across all three categories. Raw scores are reported in appendix B in table 3.

Evaluator validation (Table 2) shows consistently negative deltas for irrelevant/incorrect perturbations and low MSE for Gemini and Opus, with larger deviations for Qwen. Deltas between expected and actual changes in score per summary and section are shown in Appendix C in figure 2, 3 and 4.

8 Discussion

For the composite score, we use the geometric mean, as taking the arithmetic mean would reward degenerate outputs: For example, an empty response would get a score of 66% (full faithfulness and conciseness, zero completeness). Similarly, if a model just copied the transcript, it would score 100% on completeness and faithfulness and a below-average, but non-zero score for conciseness. The geometric mean addresses this by penalizing low scores on any single metric more heavily, thereby rewarding balanced outputs.

The results (table 1) show larger and more recent models outperforming smaller, older models - an expected outcome suggesting that the evaluation metrics correlate with general model capabilities.

Unified and Split Approach The comparison between unified and split generation supports the hypothesis that the unified approach enables higher scores but requires larger context windows and more sophisticated reasoning to handle the increased instruction complexity. High-performing models such as Claude Opus 4.5 and DeepSeek V3.2 benefit from access to the full context when generating from a unified prompt, whereas lower-performing models achieve better results with the simpler split approach. This effect is most apparent with GPT 5 Nano, which collapses under the unified approach, dropping from a 49.7% composite score with split generation to just 9.1% with unified generation.

Evaluator validation The perturbation analysis (table 2) shows that evaluator responses generally move in the expected direction, showing that evaluators are a useful way of ranking generated summaries. However, correlation is not perfect. Especially the change in completeness when omitting facts is deviating strongly from the expected baseline and is not moving at all in 46% to 36% of sections. In the other categories the scores change in the expected direction in approximately 70% to 80% of sections. There are multiple possible reasons for this: The baseline itself is approximate. It assumes two removed sentences correspond to two key-facts and that added sentences are always unfaithful or unaligned, so deviations reflect both evaluator error and simplifying assumptions. Due to human error when perturbing the originals, sentences not covering a key fact might be removed or it might be overlooked that there were multiple sentences covering a key fact, so removing one does

not change anything. Furthermore, models might infer the fact from the other sentences - this would then be a mistake in the evaluators.

Gemini 2.0 Flash performs best in evaluating conciseness and faithfulness while GPT 5 Nano performs best when evaluating completeness. It is likely that the performance of all models can be improved by experimenting with other prompts. For example, Qwen tracks the expected changes closely in most cases but has occasional outliers that inflate its MSE, particularly for omission perturbations. Through careful prompting or more sophisticated model selection (e.g. use Qwen as evaluator unless it's score deviates from the average evaluator score by more than some threshold) evaluator performance can likely be improved.

Weighted Evaluator Aggregation Given the variability across evaluators, a weighted aggregation could be applied when combining evaluator scores. Weights could be derived from perturbation-based MSE or, when available, from correlation with human judgment. This would allow selecting the best section from multiple generators using evaluator-weighted scores and improve expected performance during deployment.

Cross-Lingual Generalization Results in the Dutch subsample mostly align with the English benchmarks. On average, evaluations are slightly worse. These differences might be attributable to (i) large language models being mostly trained on English text and therefore performing slightly better in English, (ii) Claude Opus missing as an evaluator in the Dutch sample and (iii) expected variance between runs.

Limitations The framework was only evaluated on a limited dataset of 10 consultations. The English translation was done automatically using DeepL and could not be manually verified by us as we have no Dutch language skills. While translators are very accurate, this introduces a small chance for unwanted changes in the transcripts, summaries or key fact lists. Our framework for evaluation, which would be required to use the weighted aggregation to pick the best generated version also depends on the key fact lists being provided which is unrealistic in real-world deployments. When validating evaluators, the perturbations were performed manually by the authors. While we carefully checked our work, we do not have domain knowledge in medicine, therefore there is a small chance that errors occurred. Additionally, while the perturbation analysis is a good validity test, it does not replace more detailed analysis by human experts, as proposed in section 5.

Future Work There are several directions to expand on this work in the future.

First, we plan to evaluate additional generation strategies, including alternative prompts (e.g., whether providing example SOAPs improves quality) and further decomposition for smaller models, such as extracting key-facts before generating the final note. Additionally, this key-fact list could then also be used to pick the best generated summary based on weighted evaluator scores without relying on key-fact lists being provided. Especially the currently low score in conciseness should be analyzed further and, if possible, improved through better prompting.

We also plan to analyze run-to-run variance for both generators and evaluators when given the same or very similar inputs (e.g., only changing grammar) to evaluate consistency and which factors influence it.

9 Conclusion

We presented a framework for generating SOAP-notes from GP consultation transcripts, comparing unified and split generation approaches across various LLMs. To evaluate output quality, we adapted the FineSurE framework. We test the framework both using evaluation per section as well as unified for the whole summary and enhance prompts to better fit the subject. Our results show that larger models benefit from unified generation, achieving composite scores up to 77.6%, while smaller models perform better with the simpler split approach. Faithfulness scores were consistently high, while conciseness proved more challenging across all models (43–58%). We test validity of our evaluators by analyzing their score changes in reaction to perturbations of the summaries - removing key facts, adding unnecessary facts and adding wrong sentences. We additionally proposed a human evaluation protocol to enable future validation of automatic-expert alignment. This could then also be used, together with the perturbation tests, to create a weighted aggregator that chooses the best generated summary across all models.

References

Asma Ben Abacha, Wen-wai Yim, George Michalopoulos, and Thomas Lin. 2023. An investigation of evaluation methods in automatic medical note generation. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 2575–2588.

Ashfaq Liu Neubig Bhandari, Gour. 2020. Realsumm: Re-evaluating evaluation in summarization.

Knut De Swert. 2012. Calculating inter-coder reliability in media content analysis using krippendorff's alpha. *Center for Politics and Communication*, 15(1-15):3.

Klaus Krippendorff. 2011. Computing krippendorff's alpha-reliability.

Yuho Lee, Taewon Yun, Jason Cai, Hang Su, and Hwanjun Song. 2024. Unisumeval: Towards unified, fine-grained, multi-dimensional summarization evaluation for llms. *arXiv preprint arXiv:2409.19898*.

Hui Yi Leong, Yifan Gao, and Shuai Ji. 2024. A gen ai framework for medical note generation. In *2024 6th international conference on artificial intelligence and computer applications (ICAICA)*, pages 423–429. IEEE.

Artidoro Pagnoni, Vidhisha Balachandran, and Yulia Tsvetkov. 2021. Understanding factuality in abstractive summarization with frank: A benchmark for factuality metrics. *arXiv preprint arXiv:2104.13346*.

Hwanjun Song, Hang Su, Igor Shalyminov, Jason Cai, and Saab Mansour. 2024. Finesure: Fine-grained summarization evaluation using llms. *arXiv preprint arXiv:2407.00908*.

Nahathai Wongpakaran, Tinakon Wongpakaran, Danny Wedding, and Kilem L Gwet. 2013. A comparison of cohen's kappa and gwet's ac1 when calculating inter-rater reliability coefficients: a study conducted with personality disorder samples. *BMC medical research methodology*, 13(1):61.

A Human Evaluation Survey

General Instructions

Read the consultation script. For each consultation, hereinafter referred to as the source record, (multiple) SOAP-notes are presented. For each section of the SOAP-notes, evaluate:

1. **Faithfulness (sentence-level)**: whether each sentence is factually supported by the source record.
2. **Completeness (key-fact level)**: whether all relevant key-facts appear in the generated note.
3. **Conciseness (alignment level)**: whether each sentence aligns to at least one key-fact.

Annotators have access to the source clinical record and its predefined key-fact list.

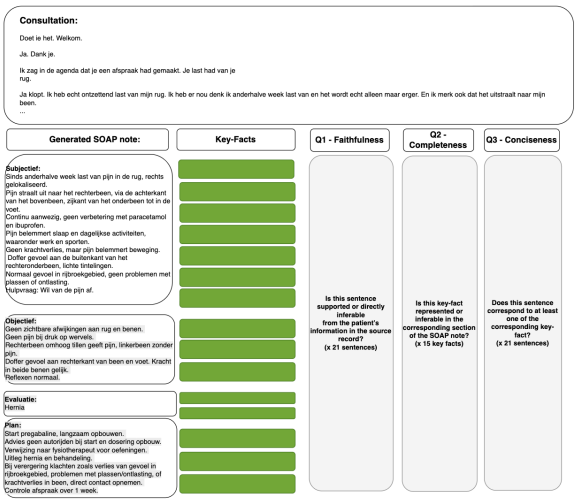

Figure 1: Structure of the Survey

Subjective

Faithfulness (Sentence-Level)

For each sentence in the *Subjective* section:

1. Is this sentence supported or directly inferable from the patient's information in the source record?
   - ( ) Yes
   - ( ) No
2. If "No", select the error type:
   - ( ) Extrinsic (information not in source)
   - ( ) Incorrect entity
   - ( ) Incorrect relation
   - ( ) Incorrect temporal detail
   - ( ) Incorrect predicate/claim
   - ( ) Coreference error
   - ( ) Other (specify)

Completeness (Key-Fact Level)

For each key-fact belonging to the *Subjective* section:

1. Is this key-fact represented or inferable in the Subjective section of the SOAP-note?
   - ( ) Yes
   - ( ) No

Conciseness (Key-Fact Alignment)

For each sentence in the *Subjective* section:

1. Does this sentence correspond to at least one Subjective key-fact?
   - ( ) Yes (list key-fact IDs)
   - ( ) No

Objective

Faithfulness (Sentence-Level)

For each sentence in the *Objective* section:

1. Is this sentence supported by exam findings, vitals, labs, or imaging in the source record?
   - ( ) Yes
   - ( ) No
2. If "No", choose the error type:
   - ( ) Extrinsic information
   - ( ) Incorrect entity
   - ( ) Incorrect relation
   - ( ) Incorrect temporal information
   - ( ) Incorrect predicate
   - ( ) Coreference error
   - ( ) Other (specify)

Completeness (Key-Fact Level)

For each key-fact in the *Objective* key-fact list:

1. Is this key-fact present or inferable in the Objective section of the SOAP-note?
   - ( ) Yes
   - ( ) No

Conciseness (Key-Fact Alignment)

For each sentence in the *Objective* section:

1. Does this sentence align with one or more Objective key-facts?
   - ( ) Yes (list key-fact IDs)
   - ( ) No

Assessment

Faithfulness (Sentence-Level)

For each sentence in the *Assessment* section:

1. Is this diagnostic or interpretative statement supported by the source record?
   - ( ) Yes
   - ( ) No
2. If "No", choose the error type:
   - ( ) Unsupported clinical inference
   - ( ) Incorrect diagnosis
   - ( ) Incorrect relation or causal link
   - ( ) Incorrect temporal sequence
   - ( ) Extrinsic (not in source)
   - ( ) Other (specify)

Completeness (Key-Fact Level)

For each key-fact in the *Assessment* key-fact list:

1. Is this key-fact represented or inferable in the Assessment section of the SOAP-note?
   - ( ) Yes
   - ( ) No

Conciseness (Key-Fact Alignment)

For each sentence in the *Assessment* section:

1. Does this sentence align to one or more Assessment key-facts?
   - ( ) Yes (list key-fact IDs)
   - ( ) No

Plan

Faithfulness (Sentence-Level)

For each sentence in the *Plan* section:

1. Is this plan item (medication, test, referral, follow-up) supported or inferable from the source record?
   - ( ) Yes
   - ( ) No
2. If "No", select the error type:
   - ( ) Extrinsic plan item
   - ( ) Incorrect medication or dosage
   - ( ) Incorrect investigation or test
   - ( ) Incorrect rationale
   - ( ) Incorrect temporal information
   - ( ) Other (specify)

Completeness (Key-Fact Level)

For each key-fact in the *Plan* key-fact list:

1. Is this plan key-fact included or inferable in the Plan section of the SOAP-note?
   - ( ) Yes
   - ( ) No

Conciseness (Key-Fact Alignment)

For each sentence in the *Plan* section:

1. Does this sentence map to one or more Plan key-facts?
   - ( ) Yes (list key-fact IDs)
   - ( ) No

B Dutch Subset Results

Table 3: Dutch subset results (section-level). Means aggregated across evaluators.

Model (Dutch)	Faithfulness	Completeness	Conciseness	Composite
gemini-2-0-flash-001_unified	97.1±1.6	84.7±7.8	56.1±11.8	77.0±7.5
gemini-2-0-flash-001_sectional	97.6±1.5	85.0±8.1	43.0±11.6	70.4±7.5
qwen3-next-80b-a3b-thinking_unified	46.9±49.5	36.1±38.2	26.7±29.6	35.4±37.6
qwen3-next-80b-a3b-thinking_sectional	89.2±8.0	40.5±28.2	38.5±17.5	49.0±17.3
gpt-5-nano_unified	8.9±28.1	2.6±8.2	5.0±15.8	4.9±15.4
gpt-5-nano_sectional	84.5±11.1	27.2±21.1	40.0±21.7	42.5±21.5

C Evaluator Validation

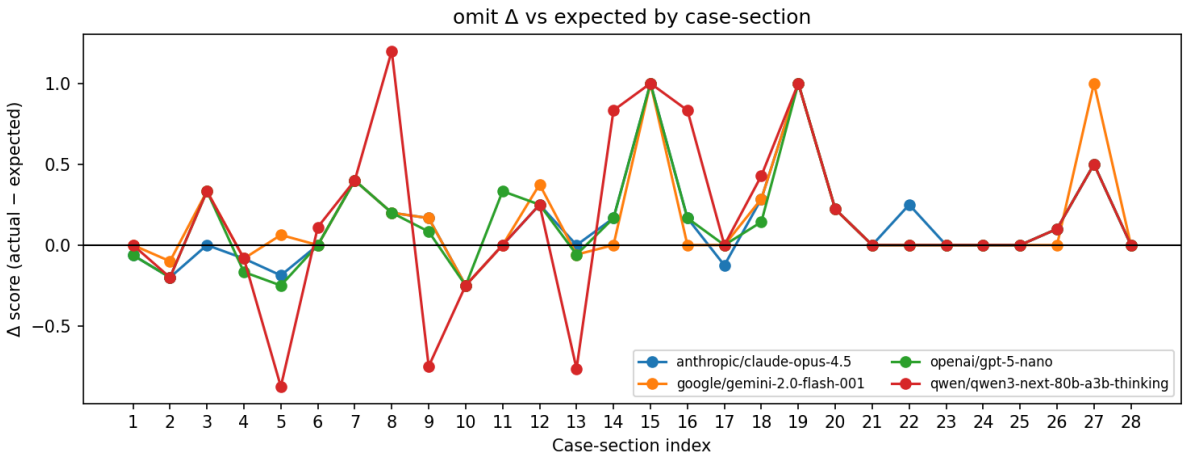

Figure 2: Case-section deltas for omission perturbations (completeness).

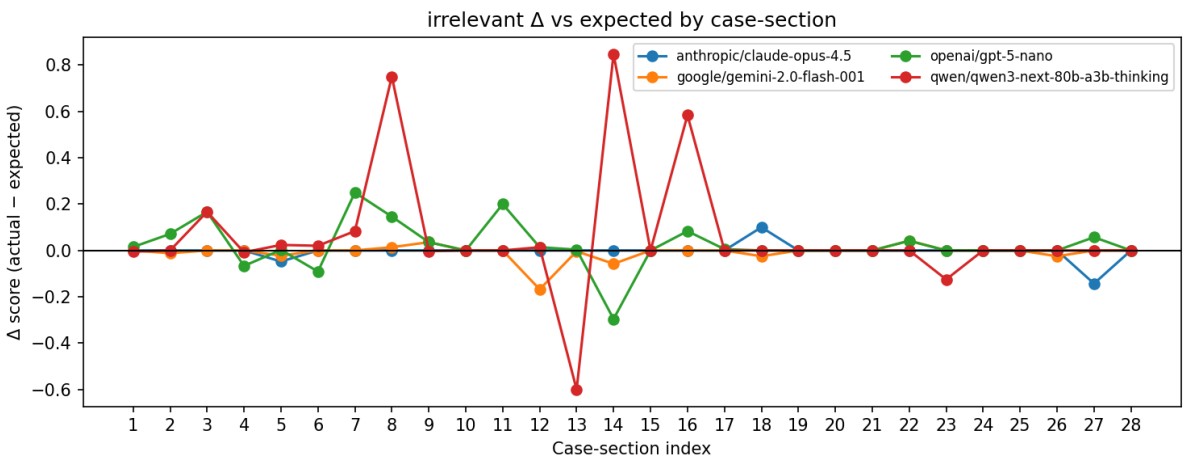

Figure 3: Case-section deltas for irrelevant additions (conciseness).

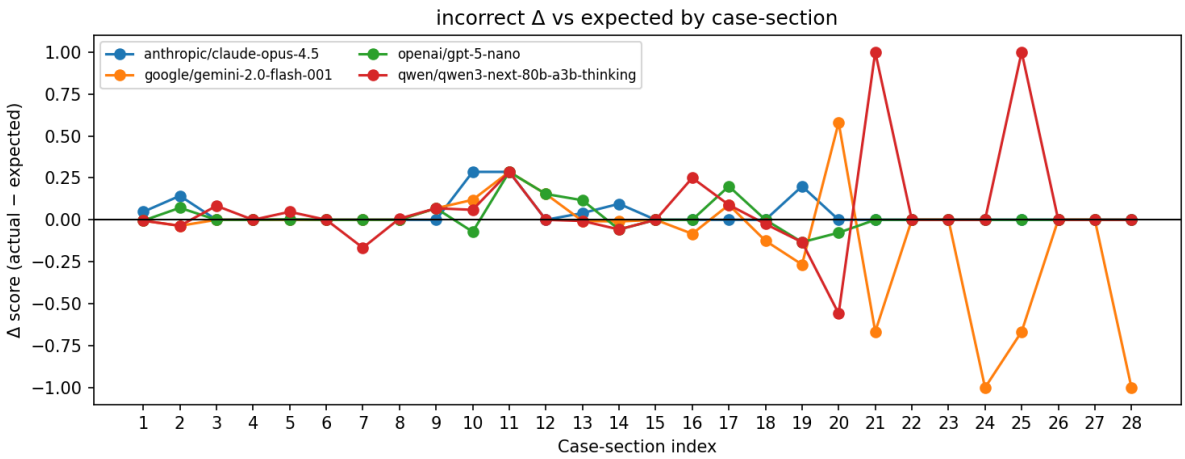

Figure 4: Case-section deltas for incorrect additions (faithfulness).